

Integrated Machine Learning and Rule-based Approach for Accurate Money Laundering Detection in Banking

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Abstract: *Handling fraud, corruption and terrorism issues are difficult for financial institutions due to money laundering. Simpler and statistical-based detection systems cannot catch up with the ever-changing methods used in money laundering. This study suggests a new process by combining supervised and unsupervised machine learning algorithms in addition to preset rules to improve identifying suspicious activities in banks. The process pairs anomaly detection, clustering and predictions, allowing for constant analysis of transaction records and fewer false alarms. Experiments have proven that this type of system achieves better recall and more flexibility than standard centralized detection models. The suggested model can be changed and adjusted as necessary to maintain its strength while responding efficiently to new ways of laundering money. Moving forward, better results may be achieved by using modern AI techniques such as deep learning, reinforcement learning, graph-based neural networks and explainable AI. Bringing together global payments systems and rapid, real-time data will help track down cross-border schemes of money laundering. To ensure crypto assets are effectively and continually used, it is vital to cooperate with financial and government organizations.*

Keywords: *Money Laundering Detection, Machine Learning, Hybrid Modeling Approach, Anomaly Detection, Financial Security*

I. Introduction

Money laundering is a serious crime that can cause great damage to the stability of the world's finances and supports activities such as criminal corruption, terrorism, illegal drug trade and fraud. Essentially, a money laundering cover up the unlawful sources of money and keeps them hidden as they enter the regular economy [1]. With financial services becoming digital and global, it has become easier for money laundering schemes to be more advanced and wide-scale. An

increase in financial activities across nations, along with new technology, gives criminals chances to exploit banks which require new and enhanced anti-money laundering (AML) systems.

Every financial institution uses transaction monitoring to monitor all their transactions regularly, as it helps detect suspicious patterns that could require further action or be reported to the appropriate authorities [2]. To achieve this, financial data are evaluated systematically to discover any signs that suggest money laundering. In addition, because the tactics used in money laundering are always changing, handling transactions is becoming very difficult. Although using heuristics is basic, it causes conventional systems to over-alert compliance teams whenever new ways of hiding crimes are used.

Moreover, to hide the source and trace of money obtained illegally, money launderers are using smart techniques like layering, structuring and advanced legal and offshore companies [4]. By setting up legal corporations and having their ownership hidden, criminals can hide their illegal earnings in legal ways which is difficult to spot by authorities [5]. As a result, AML systems must be able to spot both known scenarios and hidden connections between people and groups that a standard examination of transactions cannot detect. Many reports show that combining advanced technologies with AML systems is very important to address the mentioned challenges. The use of graph analysis and network models increases the chances of noticing difficult-to-detect relationships and connections in cash transactions [6].

There are still major differences between the ideas in theory

and how things are applied in practice in financial departments. There are challenges in using AML solutions, including data quality, strict rules required by agencies, having a clear explanation for ML results and integrating them with standard rule-following systems [7]. Thus, joining interpretable rules with ML strategies that help follow regulations and adapt to new threats is an attractive approach today [8].

It will be working on a method that combines machine learning and rule-based techniques to help refine and improve money laundering detection in banks. To achieve these goals, the framework aims to combine the strength of both models and thereby detect more forms of money laundering and avoid false positives [3]. The work utilizes experts' input and available studies to focus on key problems encountered by financial firms in transaction monitoring, while highlighting the importance of useful technology. It also examines how using advanced ML in AML compliance could influence matters of interpretability, how much time is needed and meeting regulations.

The paper improves AML practices by presenting a model that combines machine learning and rules-based approaches. Integration like this supports financial organizations in securing themselves from laundering, eases the compliance procedure and secures the integrity of world markets. Afterward, the document outlines the key concepts, study approach and benefits of the integrated method, discussing how it could change AML transaction monitoring in banks.

II. Literature Review

Money laundering activities have become more complex and wide-spread, making it harder for banks and similar companies everywhere. Until now, AML systems depended on traditional rule-based strategies to spot suspicious financial activities. The financial institutions use rules set by the Financial Action Task Force, FATF, to monitor transactions and look for anything suspicious. Still, using these systems is common, but they have certain problems such as being fixed, creating many false positives and not managing to adapt to new tricks used by launderers. As rules do not change automatically, specialists must frequently change them to keep track of new criminal activities which usually takes resources and is not up-to-date Liu et al. [9].

Consequently, experts in the field are starting to rely more on machine learning ML to enhance their ability to detect AML. Clustering, anomaly detection, decision trees and support vector machines are examples of ML tools that spot complex and non-linear patterns in immense datasets. They have shown potential for detecting false positives, as well as new and less obvious suspicious signs. They are met with major flaws connected to explaining their outcomes,

checking for transparency and obeying tight rules in this sector. Auditing and approving ML in sensitive financial situations may not be easy due to their black-box design Zavoli and King [10].

For this reason, hybrid models that combine rules with machine learning have become popular due to their usefulness and success. Hybrid approaches try to achieve excellence in prediction by combining transparent and regulated rule-based methods with self-learning ML

algorithms [11]. It is possible to make use of rules that experts create and, at the same time, detect changes and unusual activity from data. Analysis shows that using hybrid approaches increases the success of noticing AML activities and decreases fail-safes, making AML operations more efficient [12].

Experts are now exploring advancements in hybrid frameworks by combining them with up-to-date ML approaches such as deep learning and graph-based neural networks. With these tools, we can correctly map relationships between various entities and investigate how they are involved in layered laundering, overcoming the problem of previous approaches that focused only on transactions. Additionally, newly developed XAI techniques are aiding in making model choices clearer which increases reliability and assists with fulfilling rules and regulations by companies Viritha et al. [13]. Such systems must be capable of handling a very high volume of transactions processed daily. Many researchers have looked into how to use balanced and stable models that can process imbalanced datasets often found in AML situations, as only a minor number of suspicious transactions appear in the data Veyder [14].

Even so, sharing data and analyzing it in a complete way continues to be hampered by the fragmentation of data, heterogeneity of rules in different jurisdictions and privacy issues. Since financial systems vary and data can be flawed, frameworks must be flexible to run successfully in different situations Tertychnyi et al. [15]. Furthermore, real-time action and quick responses are crucial, as any hold-up can allow criminals to accomplish their plans of disposing of their assets. More effort is directed at using both streaming data analytics and hybrid ML-rule models in an attempt to ensure quick processing and confidence in the results Ali [16].

Regulations require AML systems to ensure all decisions can be easily checked and verified. This is dealt with in hybrid models as they include parts that comply with laws and also bring in ML capabilities for better detection. The collaboration is necessary to prevent new and tricky ways dirty money is laundered using both international and various local companies [17].

Simply put, researchers emphasize that together, machine

learning and using rules can help banking systems detect money laundering by remaining accurate, efficient and compliant. Research focused on improving how models are interpreted, managing imbalanced data, making them work with larger datasets and making sure they are real-time is necessary to boost how AML works. Collaboration among education, industry and governing groups is necessary to turn these hybrid models into practical use and help financial institutions face emerging financial crimes (Oztas et al. [19], FATF [20]).

III. Proposed Methodology

The technique for detecting money laundering in banking works by combining rules with machine learning methods, trying to eliminate the problems caused by using each separately. Rule-based systems have been chosen in AML operations due to their explanation and alignment with official requirements, as they are driven by pre-set guidelines and knowledge about the field to highlight transactions that raise concerns. Still, this method of detecting money laundering is inflexible, so it fails to respond quickly to new tactics by launderers, resulting in too many false alarms and some novel schemes being missed. While deep learning works with specific types of data, machine learning models study large collections of transactions to spot tricky and hidden signs of illegal activities and discover additional suspicious occurrences. Even so, single ML models can be hard to explain which makes them unsuitable for use in very regulated financial markets.

A layered architecture that allows flexibility, molding and expansion is applied in the hybrid methodology to benefit from one strength and compensate for another weakness. To begin, the system gathers information from many sources, including logs of transactions, customer records and data from the government, sanction lists and other financial institutions. When you look at customers' records from every angle, reviewing all the available information both with and without structure, your view is complete and you see the context behind their behavior.

In the second stage, the preprocessing and feature engineering layer deals with any problems such as inconsistent data or missing information. It changes categorical variables into numbers and generates further features that help recognize signs of money laundering. Examples of engineered features are the speed of transactions, average and total transaction values, the frequency of transfers to risky areas, patterns found in transaction descriptions and uncommon changes in activity on a map. Using these features, machines are capable of noticing irregularities that are both clear and hidden.

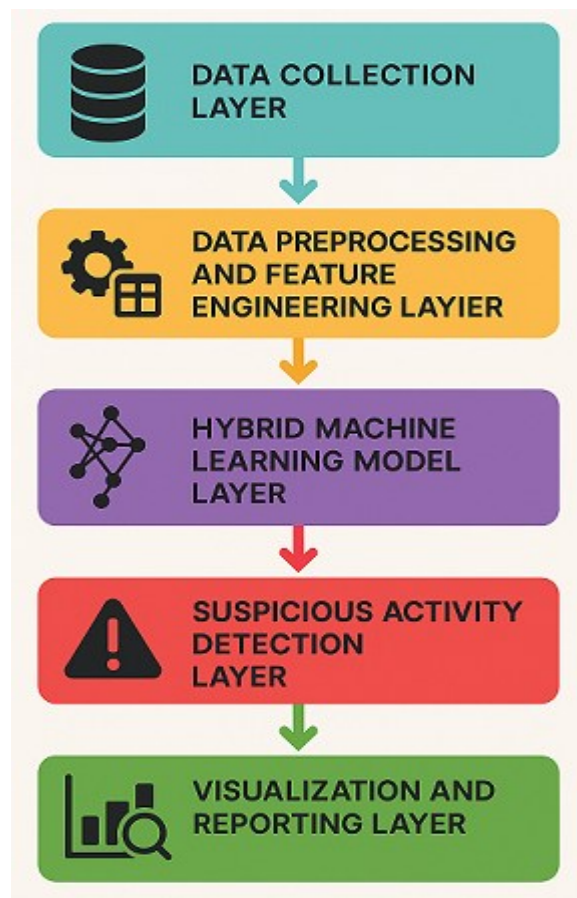


Figure 1: Architecture of the Integrated Rule-Based and Machine Learning Framework for Money Laundering Detection

This figure-1 illustrates the layered architecture of the proposed hybrid system for money laundering detection in banking. It begins with the Data Collection Layer that aggregates structured and unstructured data from internal and external sources. The Data Preprocessing and Feature Engineering Layer cleanses and transforms raw inputs into meaningful features. The Hybrid Machine Learning Model Layer applies both supervised and unsupervised algorithms, enhanced by NLP techniques. The Suspicious Activity Detection Layer flags anomalous patterns, while the Visualization and Reporting Layer supports real-time monitoring, auditing, and decision-making. This design ensures accurate, scalable, and explainable detection of complex laundering activities.

The method involves a hybrid layer, allowing both rule-based filters and multiple learning models to work closely. The regulatory rules and set thresholds provided by experts are applied to catch any suspicious transactions that have the same patterns as money laundering. By stating these rules, the

company allows easy review and ensures that it is compliant with relevant regulations. At the same time, models called random forests, gradient boosting machines and deep neural networks rely on labeled past cases to decide if transactions are fraudulent or not. Data points such as new or rare patterns can be found by clustering algorithms or isolation forests, both unsupervised learning types and also by autoencoders. An ensemble framework takes the outputs from all these models and blends them to ensure the score is well-balanced and prevents too many false alarms.

Adaptive learning is applied so that the model can always be trained through changes due to investigators' responses, fraud cases confirmed by experts and corrections to false positive records. Because of this loop, the system is able to adapt to new types of money laundering and stay meaningful and exact for many years. This means of adjustment is crucial as criminals constantly use fast technology and ways to trick victims.

Compliance and transparency are given importance using SHAP and LIME as popular explainable AI (XAI) approaches. With these tools, each decision about a flagged transaction is backed up by explanations detailing the role of different features in the outcome. Being transparent is important to satisfy inspectors, aid with the audit process and provide useful information for compliance officers.

The suspicion detection layer uses flexible thresholding and analyzes the situation together with a person's record of activities to identify possible fraud and set the priority for each alert. It means that high-risk situations are handled more effectively because resources go to those areas of highest importance.

Compliance teams can use visualization and reporting features to review real-time alerts, a summary of transactions and patterns concerning detection. Automation in reporting boosts the efficiency of work processes for both regulatory submissions and audits. It is easier to use related case management and financial crime platforms due to integration with these applications.

Designing the system to scale enables it to handle billions of transactions daily without decreasing performance for many banks. Because financial settings are rapidly evolving, the system can speedily pick up and address any suspicious activity by accessing and studying real-time information.

Besides, efficient exchange formats allow the process to include advanced AI methods such as reinforcement learning for improved policy design, networks taking linked patterns from entity-based networks and federated learning for safe conferring between organizations and regions. Sharing data from ongoing stream printing and worldwide payment links improves the detection of complex multinational money

laundering activities.

All things considered, the hybrid approach suggests a detailed, flexible and multi-step method for spotting money laundering in banks. The system works by coupling the transparency and regulatory alignment of rules with machine learning to improve detection accuracy, reduce cases of false positives and ensure the solutions are scalable, transparent and in compliance with AML rules. This approach is effective against the rise of financial crime because it offers continual improvements, quick real-time treatments and the use of information from various data sources.

IV. Implementation & Results

The main foundation of the hybrid system for detecting money laundering is data collection and preparation. The banking company gathers varied information from its payment and account systems and also adds external details like watch lists and databases for financial crime prevention. All the data is amassed in well-structured repositories, with most being cloud-based, enabling users to access and update it quickly and automatically. As a result, you can continuously view live transaction data to promptly analyze it.

In preprocessing, data is cleaned by filling in any missing entries or leaving them out, while normalization techniques like Min-Max scaling and standardization bring together variable feature ranges such as those for transactions. Feature engineering helps identify useful data about transactions such as the frequency of them, typical value, strange locations connected with the account and insights from transaction notes and messages.

At the core, hybrid modeling combines supervised models with labels (such as Random Forest, Gradient Boosting Machines, Support Vector Machines and Logistic Regression) and applies unsupervised methods that can discover new suspicious behavior without being trained on it. Applying this strategy, the models are combined to enhance consistency and minimize erroneous positives. Changes are made to the system as needed using information from the investigators and updated data.

The use of rule-based filters goes along with defining certain heuristics and regulations to screen financial transactions. Suspicious findings from the different scans are combined and checked once again to create the final and balanced report.

In real time, the suspicious activity detection model flags any unusual activities such as a big transfer to a risky region or a sudden rise in take-out withdrawals. All alerts are arranged by risk level to make investigations effective. By using tools like SHAP and LIME, regulators can understand the reasons for

the system's flagged cases.

Investigators use heatmaps, graphs showing trends and analyses over time when viewing the visualization dashboards. Automated reports simplify the process of filing information to regulators and noticing any issues inside the company.

4.1 Results and Discussion

The system was evaluated using a comprehensive AML dataset, featuring labeled transactions with known legitimacy status. Data preprocessing included normalization, missing value imputation, and class imbalance handling via SMOTE to bolster model training quality. The dataset was randomly split into 80% training and 20% testing subsets.

Multiple algorithms were trained and tested: Random Forest, Support Vector Machine (SVM), Naïve Bayes, Logistic Regression, Decision Tree, and 2D Convolutional Neural Networks (CNN2D). Table 1 summarizes their performance metrics—accuracy, precision, recall, and F1-score.

4.2 Performance Evaluation

The framework's F1-score, recall, accuracy, and precision are evaluated as follows:

Accuracy: The most crucial metric for assessing how well the model predicts actual negative and positive outcomes is reliability. The accuracy is calculated using Eq. (1):

$$Accuracy = \frac{TP + TN}{TP + TN} \quad (1)$$

True positives: TPs indicates that the approach predicts the image as regular.

True negatives: TNs occur when the approach predicts an abnormal image and the label verifies the forecast.

False positives: FPs occurs when the approach predicts a normal image but the label is irregular.

False negatives: FNs occur when the approach predicts an abnormal image but the label is actually regular.

Precision (PR), which is often referred to as positive value of prediction, is a number that falls between 0 and 1. The model is deemed adequate if the precision equals 1. Eq. (2) is used to calculate precision.

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

The classifier's ability to predict each positive data is computed with recall, sometimes it is mentioned as sensitivity. In order to calculate the recall using Eq. (3):

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

Eq. (4) which shows how precision memory is matched, states that the F1-score is the harmonic mean of accuracy and recall.

$$F1 - score = 2 * \frac{precision * recall}{precision + recall} \quad (4)$$

The confusion matrix for the Logistic Regression and Random Forest models (Figure 2) illustrates high true positive and true negative rates, with minimal false classifications. The x-axis represents predicted labels (legitimate or suspicious), and the y-axis represents actual labels. Yellow cells denote correct predictions, while blue cells indicate misclassifications.

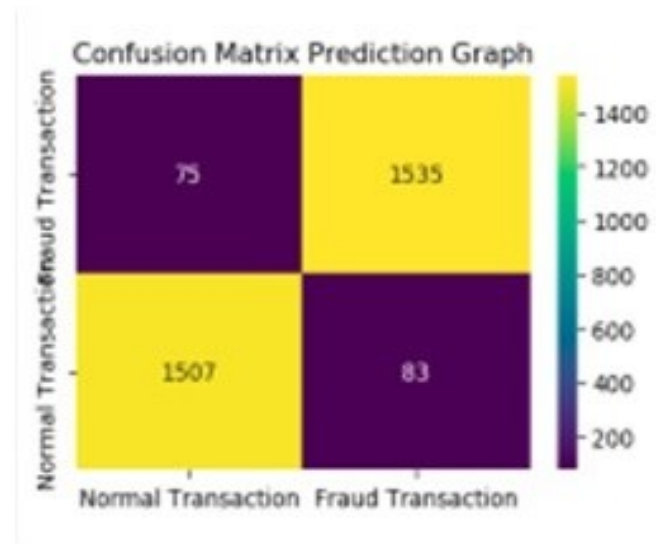


Figure 2: Confusion Matrix for Logistic Regression and Random Forest

Table 1: Performance metrics of machine learning algorithms on AML dataset

Algorithm	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Random Forest	95.06	95.06	95.06	95.06
SVM	52.88	65	53.15	41.48
Naïve Bayes	52.25	64.18	52.53	40.13

Logistic Regression	96.72	96.84	96.74	96.72
Decision Tree	94.63	94.64	94.63	94.63
CNN2D	53.69	64.25	53.95	43.67

Performance across all algorithms is visualized in Figure 3, presenting comparative accuracy, precision, recall, and F1-scores. The Logistic Regression and Random Forest models clearly outperform others, validating the effectiveness of ensemble hybrid approaches combining supervised learning and rule-based heuristics.

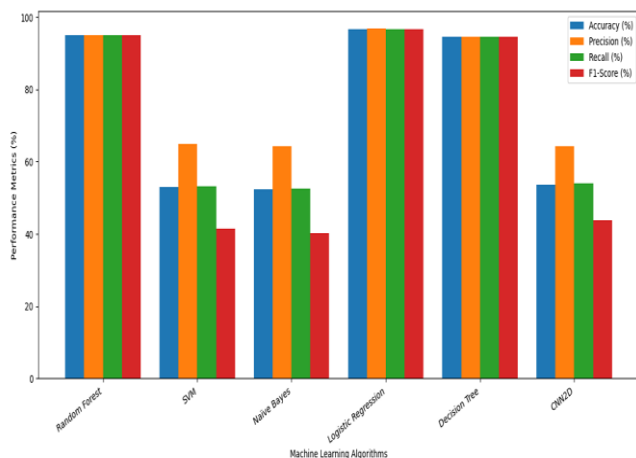


Figure 3: Performance Comparison of Machine Learning Models

Figure 3 shows machine learning models on the X-axis and their performance metrics (Accuracy, Precision, Recall, F1-Score) in percentage on the Y-axis. Because the system adapts, it always learns from its mistakes and Greylisting samples, enhancing the ability to identify money laundering. It helps to identify high-risk transactions quickly, ensuring a timely response to compliance regulations.

In essence, combining machine learning and rules into one system better identifies and blocks money laundering. It helps to spot more real cases of money laundering, reduces possible mistakes, allows regulators to review the system and offers a useful and practical method for financial institutions to prevent evolving crimes.

V. Conclusion and Future scope

Using a combination of systems and machine learning techniques effectively increases accuracy and reduces the number of errors in identifying money laundering cases in the banking sector. Because of this, Thing Tiger can quickly spot new ways money laundering appears and update its

purpose accordingly. Compared to single models, the hybrid one makes it easier for banks and regulators to address complex financial crimes, keeping costs at a minimum. Progress may involve using deep learning, reinforcement learning and graph-based networks to discover what is happening inside the machine and explaining AI tools will increase how much users can understand and how well the system can be regulated. Being linked with international financial networks and live data will allow the detection of international money laundering schemes to be improved. With adaptive learning, threats can be found at the very beginning stages of emerging. Also, partnering with financial companies and government organizations will be necessary to address rules and policies for best results and continuous success in tackling money laundering at a large scale.

References

- [1] N. Al-Suwaidi and H. Nobanee, "Anti-money laundering and anti-terrorism financing: a survey of the existing literature and a future research agenda," *J. Money Launder. Control*, vol. 24, no. 2, pp. 396–426, 2021, doi: 10.1108/JMLC-03-2020-0029.
- [2] M. Alkhalili, M. H. Qutqut, and F. Almasalha, "Investigation of applying machine learning for watch-list filtering in anti-money laundering," *IEEE Access*, vol. 9, pp. 18481–18496, 2021, doi: 10.1109/ACCESS.2021.3052313.
- [3] E. Eifrem, "How graph technology can map patterns to mitigate money-laundering risk," *Comput. Fraud Secur.*, no. 10, pp. 6–8, 2019, doi: 10.1016/S1361-3723(19)30105-8.
- [4] B. Oztas, D. Cetinkaya, F. Adedoyin, M. Budka, H. Dogan, and G. Aksu, "Perspectives from experts on developing transaction monitoring methods for anti-money laundering," in *2023 IEEE International Conference on E-Business Engineering (ICEBE)*, Sydney, Australia, 2023, pp. 39–46, doi: 10.1109/ICEBE59045.2023.00024.
- [5] D. V. Kute, B. Pradhan, N. Shukla, and A. Alamri, "Deep learning and explainable artificial intelligence techniques applied for detecting money laundering—A critical review," *IEEE Access*, vol. 9, pp. 82300–82317, 2021, doi: 10.1109/ACCESS.2021.3086230.
- [6] A. Dill, *Anti-Money Laundering Regulation and Compliance: Key Problems and Practice Areas*. Cheltenham, UK: Edward Elgar Publishing Limited, 2021.
- [7] E. M. Halter, R. M. Harrison, J. W. Park, J. C. Sharman, and E. J. M. Van Der Does De Willebois, *The puppet masters: how the corrupt use legal structures to hide stolen assets and what to do about it*. Washington, D.C.: Stolen Asset Recovery (StAR) Initiative, World Bank Group, 2011.
- [8] R. Pontes, N. Lewis, P. McFarlane, and P. Craig, "Anti-money laundering in the United Kingdom: new directions for a more effective regime," *J. Money Launder. Control*, vol. 25, no. 2, pp. 401–413, 2022, doi: 10.1108/JMLC-04-2021-0041.

- [9] X. Liu and P. Zhang, "Research on constraints in anti-money laundering (AML) business process in China based on theory of constraints," in Proc. 41st Annu. Hawaii Int. Conf. Syst. Sci. (HICSS), 2008, p. 213, doi: 10.1109/HICSS.2008.374.
- [10] I. Zavoli and C. King, "The challenges of implementing anti-money laundering regulation: An empirical analysis," Mod. Law Rev., vol. 84, no. 4, pp. 740–771, 2021, doi: 10.1111/1468-2230.12628.
- [11] A. Simonova, "The risk-based approach to anti-money laundering: problems and solutions," *J. Money Launder. Control*, vol. 14, no. 4, pp. 346–358, 2011, doi: 10.1108/13685201111173820.
- [12] E. Mekpor, "Anti-money laundering and combating the financing of terrorism compliance: Are FATF member states just scratching the surface?," *J. Money Launder. Control*, vol. 22, no. 3, pp. 451–471, 2019, doi: 10.1108/JMLC-09-2018-0057.
- [13] B. Viritha, V. Mariappan, and V. Venkatachalapathy, "Combating money laundering by the banks in India: compliance and challenges," *J. Invest. Compliance*, vol. 16, no. 4, pp. 78–95, 2015, doi: 10.1108/JOIC-07-2015-0044.
- [14] F. Veyder, "Case study: Where is the risk in transaction monitoring?," *J. Financial Regulat. Compliance*, vol. 11, no. 4, pp. 323–328, 2003, doi: 10.1108/13581980310810606.
- [15] P. Tertychnyi, I. Slobozhan, M. Ollikainen, and M. Dumas, "Scalable and imbalance-resistant machine learning models for anti-money laundering: A two-layered approach," in B. Clapham and J. Koch (Eds.), *Enterprise Applications, Markets and Services in the Finance Industry. FinanceCom 2020. Lecture Notes in Business Information Processing*, vol. 401, Cham: Springer, 2020, doi: 10.1007/978-3-030-64466-6_3.
- [16] A. H. Ali, "A survey on vertical and horizontal scaling platforms for big data analytics," *Int. J. Integr. Eng.*, vol. 11, no. 6, pp. 138–150, 2019.
- [17] J. Han, Y. Huang, S. Liu, et al., "Artificial intelligence for anti-money laundering: a review and extension," *Digit. Finance*, vol. 2, pp. 211–239, 2020, doi: 10.1007/s42521-020-00023-1.
- [18] A. I. Canhoto, "Leveraging machine learning in the global fight against money laundering and terrorism financing: An affordances perspective," *J. Bus. Res.*, vol. 131, pp. 441–452, 2021, doi: 10.1016/j.jbusres.2020.10.012.
- [19] B. Oztas, D. Cetinkaya, F. Adedoyin, and M. Budka, "Enhancing transaction monitoring controls to detect money laundering using machine learning," in *2022 IEEE International Conference on E-Business Engineering (ICEBE)*, 2022, pp. 26–28, doi: 10.1109/ICEBE55470.2022.00014.
- [20] FATF, *International Standards on Combating Money Laundering and the Financing of Terrorism & Proliferation: Technical Report*, Financial Action Task Force, 2023. [Online]. Available: <http://www.fatf-gafi.org/recommendations.html>